FIITJEE Students From All Programs have bagged 33 in Top 100, 77 in Top 200 and 205 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2015:
4551 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE **JEE (Advanced), 2016**

PART TEST – III

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 180**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 10)** contains 10 multiple choice questions which have one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (11 to 16) contains 3 paragraphs with each having 2 questions. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (17 – 20) contains 4 Matching Lists Type questions: Each question has four statements in LIST I & 4 statements in LIST II. The codes for lists have choices (A), (B), (C), (D) out of which only one is correct. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

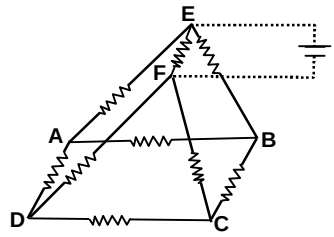
PART – I

SECTION - A

Straight Objective Type

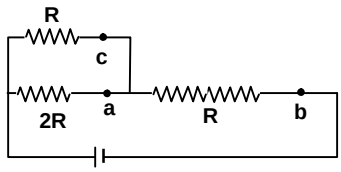
This section contains **10 multiple** choice questions numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. A prism is made of wire mesh with each side having equal resistance R. A battery of 6 V and zero resistance is connected across E and F as shown in the figure. The current in the branch AD, if R is equal to 5Ω is



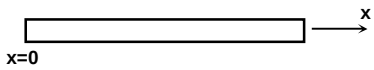
(A) 0.6A (B) 0.8 A
(C) 0.4 A (D) 2 A

2. Referring to the shown circuit, the current will be minimum in



(A) a
(B) b
(C) c
(D) same in all the branches

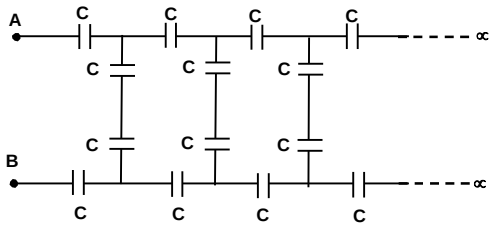
3. If along a uniform rod of length ℓ carrying current I, the voltage V changes with position x along the length of the rod such that dV/dx = – k, where k is a positive number, then the resistance of the rod is



(A) kℓ/2I (B) kℓ/I
(C) I/kℓ (D) kℓI

Space for rough work

4. The equivalent capacitance of the shown infinite network across A and B, if each capacitor has a capacitance of C, is



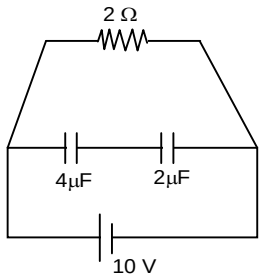
(A) $\left(\frac{\sqrt{3}-1}{4}\right)C$

(B) $\left(\frac{\sqrt{3}+1}{4}\right)C$

(C) $\left(\frac{\sqrt{5}-1}{4}\right)C$

(D) None of these

5. In the shown network, charge on 2μf capacitor in steady state would be

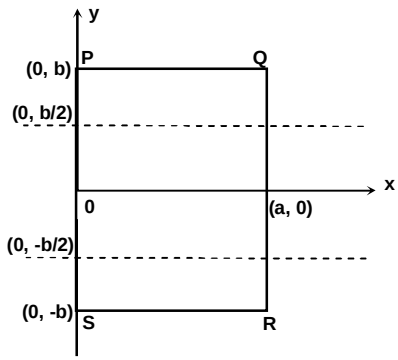


(A) 10 μC (B) 20 μC

(C) 40/3 μC (D) 40 μC

6 A uniform and constant magnetic field exists in a region as under

$$\begin{cases} B_o\hat{i} & \text{for } y > \frac{b}{2} \\ -B_o\hat{i} & \text{for } y < -\frac{b}{2} \\ \text{zero for } -\frac{b}{2} \le y \le \frac{b}{2} \end{cases}$$



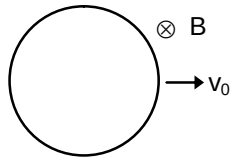
Then the current that must be passing thru the area enclosed by PQRS shown

(A) $\frac{2B_oa}{\mu_o}\left(-\hat{k}\right)$ (B) $\frac{2B_oa}{\mu_o}\left(\hat{k}\right)$

(C) $\frac{2B_ob}{\mu_o}\left(-\hat{k}\right)$ (D) $\frac{2B_ob}{\mu_o}\left(\hat{k}\right)$

Space for rough work

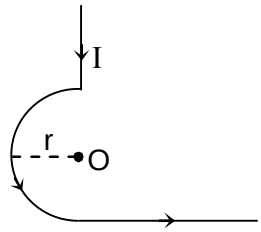
7. A uniform ring of mass m with uniformly distributed charge q along its circumference of radius R is translating with speed v_0 perpendicular to a uniform magnetic field as shown. The acceleration of the ring is (Assuming gravity free space)



(A) $2qv_0B/m$ (B) 0

(C) qv_0B/m (D) $(qv_0B)/2m$

8. In the given figure, what is the magnitude of field induction at point O?

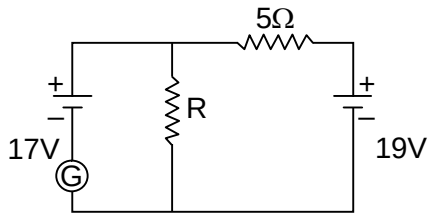


(A) $\frac{\mu_0 I}{4\pi r}$ (B) $\frac{\mu_0 I}{4r}+\frac{\mu_0 I}{2\pi r}$

(C) $\frac{\mu_0 I}{4r}+\frac{\mu_0 I}{4\pi r}$ (D) $\frac{\mu_0 I}{4r}-\frac{\mu_0 I}{4\pi r}$

9. A parallel plate capacitor is connected across a source of constant potential difference. When a dielectric plate is introduced between the two plates, then
(A) Potential difference between two plates increases
(B) Potential difference between two plates decreases
(C) Potential difference between two plates remains same
(D) Charge on the plates decreases

10. For what value of R will the current in galvanometer be zero ?
(A) 7 Ω
(B) 5 Ω
(C) 15 Ω
(D) 85/2 Ω



Space for rough work

Comprehension Type

This section contains **3 groups** of questions. Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

Paragraph for Question Nos. 11 to 12

In a region, charge density is

$$\begin{cases} \rho = \rho_o \sin\dfrac{\pi x}{L} \text{ for } 0 \le x \le L \\ = \text{zero} \quad \text{for all other values of } x \end{cases}$$

The charge density is independent of value of y and z and it extends from $-\infty < y < \infty$ and $-\infty < z < \infty$.

11. The electric field at (L/2, L/2, 0) is

(A) $\rho_o \dfrac{\pi}{2}\hat{i}$ (B) $\rho_o \dfrac{\pi}{22}\hat{i}$

(C) $\rho_o \dfrac{\pi}{\sqrt{2}L}\hat{i}$ (D) zero

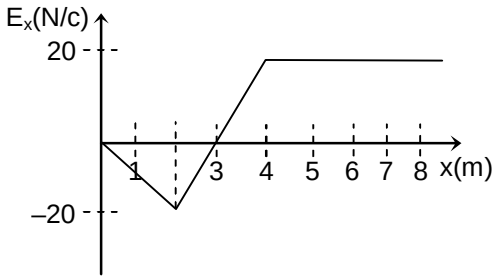
12. Equipotential surface will be

(A) Planes parallel to x-y plane (B) Planes parallel to y-z plane
(C) Planes parallel to x-z plane (D) None of these

Space for rough work

Paragraph for Question Nos. 13 to 14

A graph of the x-component of the electric field as a function of x in a region of space is shown in the figure. The y and z-components of the electric field are zero in this region. The electric potential at the origin is zero.



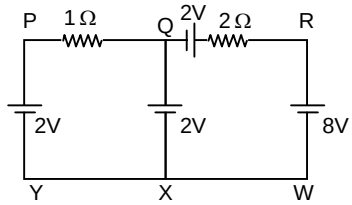
13. The electric potential at x = 2m is :
(A) 10 V (B) 20 V
(C) 30 V (D) 40 V

14. The greatest positive value of electric potential for points of the x-axis for which $0 \le x \le 6m$ is :
(A) 10 V (B) 20 V
(C) 30 V (D) 40 V

Paragraph for Question Nos. 15 to 16

For the electrical circuit shown in the figure:

Answers the following questions:



15. The current in branch PQ is
(A) zero (B) 1 A
(C) 2 A (D) 3 A

16. The current in branch QR is
(A) zero (B) 1 A
(C) 2 A (D) $\frac{2}{3}$ A

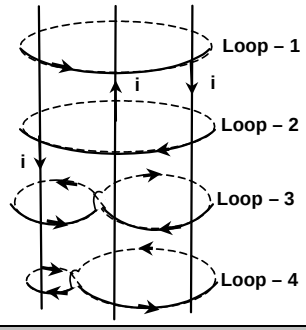
Space for rough work

(Matching list Type)

This section contains **4 multiple choice questions**. Each question has matching lists. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

17. **Read the following statements and match the following and write the correct pairs.**

Three wires are carrying same constant current in different directions. Four loops enclosing the wires in different manners are shown. The direction of $\mathbf{d}\vec{\ell}$ is shown in the figure.



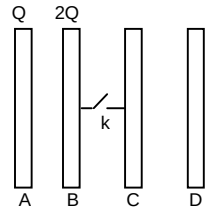
	List - I		List - II
(P)	Along closed loop – 1	(1)	$\oint \vec{B}.d\vec{\ell} = \mu_o i$
(Q)	Along closed loop – 2	(2)	$\oint \vec{B}.d\vec{\ell} = -\mu_o i$
(R)	Along closed loop – 3	(3)	$\oint \vec{B}.d\vec{\ell} = -3\mu_o i$
(S)	Along closed loop – 4	(4)	$\oint \vec{B}.d\vec{\ell} = 3\mu_o i$

Codes:

	P	Q	R	S
(A)	2	1	2	1
(B)	1	4	2	3
(C)	4	3	1	2
(D)	2	3	4	1

Space for rough work

18. *Read the following and write the correct pairs.*
Initially, the two plates (A and B) are given charges as shown. Find the final charges on the plates after the key k is closed. All plates are conducting, parallel and of infinite length and breadth.

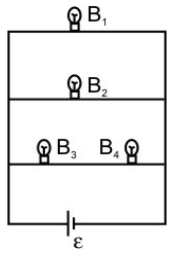


	List - I		List - II
(P)	Final charge on plate A	(1)	Q/2
(Q)	Final charge on plate B	(2)	Q
(R)	Final charge on plate C	(3)	0
(S)	Final charge on plate D	(4)	3Q/2

Codes:

	P	Q	R	S
(A)	2	3	1	4
(B)	1	4	2	3
(C)	4	3	1	2
(D)	2	1	4	3

19. Ratings of bulbs are as follows :
B_1 (100w, 200v) ; B_2 (50w, 200v) ; B_3 (100w, 200v) ; B_4 (200w, 200v)



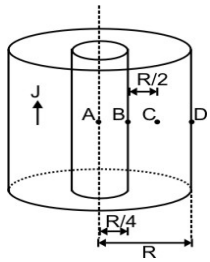
	List – I		List - II
(P)	Current drawn by B_1 are	(1)	Maximum
(Q)	Intensity of B_4 and are B_3	(2)	Minimum
(R)	Intensity of B_1	(3)	Same
(S)	Current drawn by B_2	(4)	Different

Codes:

	P	Q	R	S
(A)	2	1	2	1
(B)	1	4	1	2
(C)	4	3	1	2
(D)	2	3	4	1

Space for rough work

20. From an infinitely long cylinder of radius R, a cylinder of radius R/4 is removed and current density in the remaining portion is J as shown in the adjoining figure. BC = R/2. Match the entries of column I with the entries of column II.



List – I		List - II	
(P)	$\vec{B}$ at A is	(1)	Zero
(Q)	$\vec{B}$ at B is	(2)	$\frac{\mu_0 JR}{3}$
(R)	$\vec{B}$ at C is	(3)	$\frac{5\mu_0 JR}{3}$
(S)	$\vec{B}$ at D is	(4)	$\frac{15\mu_0 JR}{32}$

Codes:

	P	Q	R	S
(A)	2	1	2	1
(B)	1	4	2	3
(C)	1	1	2	4
(D)	2	3	4	1

Space for rough work

Chemistry

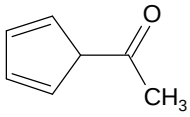
PART – II

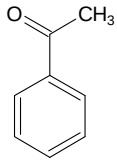
SECTION - A

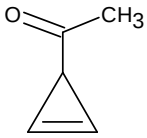
Straight Objective Type

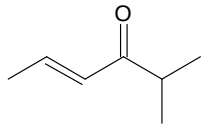
This section contains **10 multiple** choice questions numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. Which of the following will be most acidic?

(A) 

(B) 

(C) 

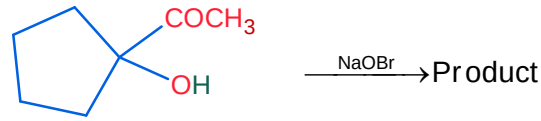
(D) 

2. 1 - propoxy butane can be prepared by
   (A) treating 1 – bromopropane with sodium 1 – propoxide
   (B) heating with 3 – pentanol with conc. H_2SO_4 at 473 K.
   (C) treating with 2 – bromopropane with sodium isopropoxide
   (D) heating 1 – propanol with conc. H_2SO_4 at 443 K

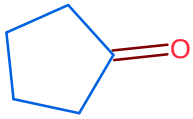
3. Bromination of aniline gives 2, 4, 6 – tribromoaniline where as the nitration of aniline with mixed acids gives m – nitroaniline. In this case nitration, the m – derivative is formed because
   (A) In the presence of strong acids the amino group is protonated to $-\overset{\oplus}{N}H_3$ which is m – orienting
   (B) m – nitro aniline is thermodynamically more stable than the ortho and para-isomers
   (C) nitro groups can not enter ortho and para-positions due to steric factor
   (D) the mechanism of bromination and nitration are different

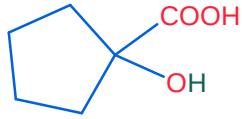
4. The major product formed by the reaction of a anisole with tert-butyl alcohol is
   (A) 1- methoxycyclohexa – 1,4 diene
   (B) 2 – methoxycyclohexa -1,3 diene
   (C) 1- methoxycyclohexa – 1,3 diene
   (D) 3- methoxycyclohexa – 1,4 diene

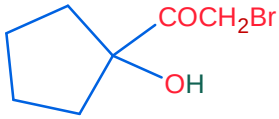
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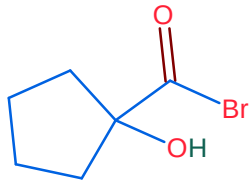
5. 

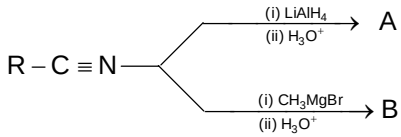
The product is

(A) 

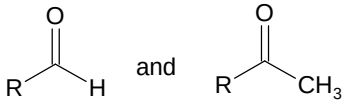
(B) 

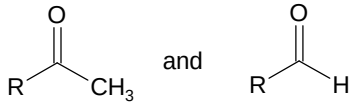
(C) 

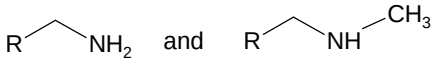
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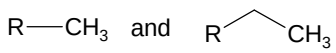
6. 

A and B are

(A) 

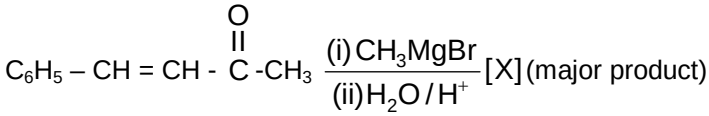
(B) 

(C) 

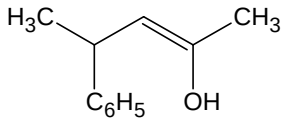
(D) 

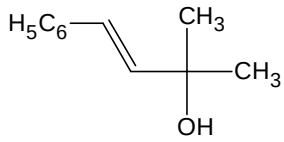
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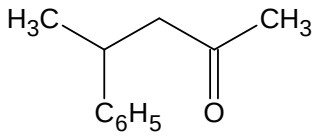
7. In the reaction sequence

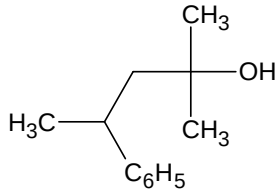


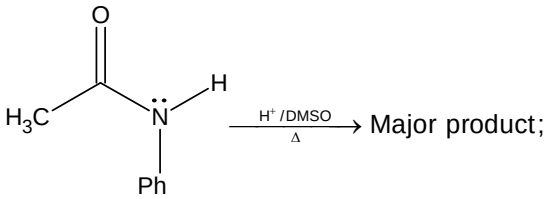
X will be

(A) 

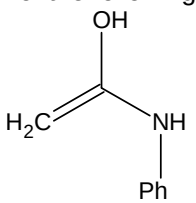
(B) 

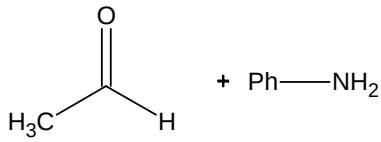
(C) 

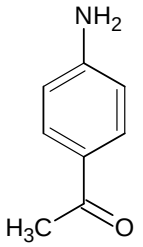
(D) 

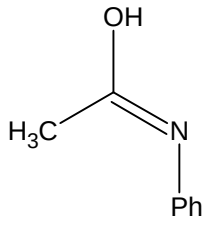
8. 

Which of the following is major product?

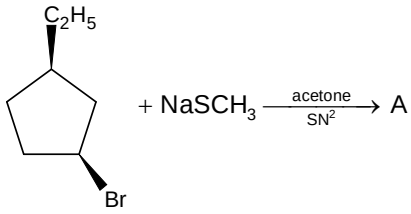
(A) 

(B) 

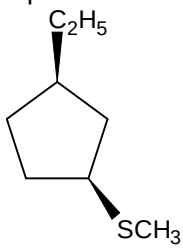
(C) 

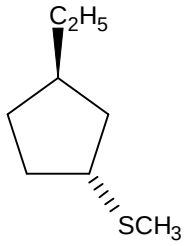
(D) 

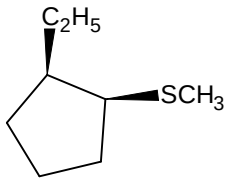
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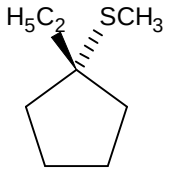
9. 

The compound A is –

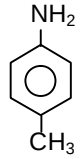
(A) 

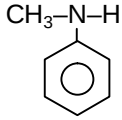
(B) 

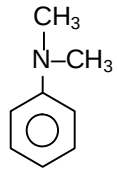
(C) 

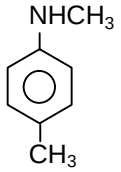
(D) 

10. Which of the following amine will show carbylamine reaction?

(A) 

(B) 

(C) 

(D) 

Space for rough work

Comprehension Type

This section contains **3 groups** of questions. Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Question Nos. 11 to 12

Aldehydes without α - hydrogen undergoes Cannizzaro reaction in the presence of conc. Alkali. Two molecule undergoes self oxidation and reduction to yield a mixture of alcohol and salt of carboxylic acid. Two different aldehydes without α hydrogen undergoes crossed Cannizzaro reaction. The rate of reaction is decided by the ease with which H^- is transferred.

11. $HCHO + PhCHO \xrightarrow{50\%\ NaOH} P + Q$

P and Q are

(A) HCOOH and PhCOONa
(B) HCOONa and PhCOOH
(C) CH_3OH and PhCOONa
(D) HCOONa and $PhCH_2OH$

12. $(H_3C)_3C-\overset{O}{\overset{\|}{C}}-\overset{O}{\overset{\|}{C}}-H \xrightarrow{conc.\ NaOH} Product(s)$

Choose the correct statements among the following statement:

(A) product are $(H_3C)_3C-\overset{O}{\overset{\|}{C}}-COONa$ and $(H_3C)_3C-\overset{O}{\overset{\|}{C}}-CH_2OH$

(B) Product is $(H_3C)_3C-\overset{OH}{\underset{|}{C}}-\overset{O}{\overset{\|}{C}}-ONa$ and reaction in disproportionation reaction

(C) Product is $(H_3C)_3C-\overset{OH}{\underset{|}{C}}-\overset{O}{\overset{\|}{C}}-ONa$ but is not disproportionation reaction

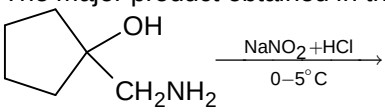
(D) Product are $(H_3C)_3C-\overset{OH}{\underset{|}{C}}-\overset{O}{\overset{\|}{C}}-H$ and $(H_3C)_3C-\overset{O}{\overset{\|}{C}}-CH_2OH$

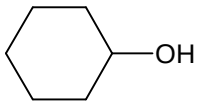
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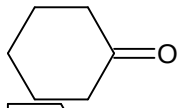
Paragraph for Question Nos. 13 to 14

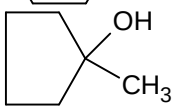
Aliphatic primary amines react with nitrous acid ($NaNO_2$ + HCl) form alcohol as the major product. In addition to alcohol, alkenes and alkyl halide are also formed as minor products. Nitrous acid also react with secondary and tertiary amine, nature of reaction depends whether amine is aliphatic or aromatic.

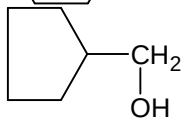
13. The major product obtained in the following reaction will be



(A) 

(B) 

(C) 

(D) 

14. Which of the following product(s) will be obtained when n-propyl amine is treated with $NaNO_2$ and HCl?

(A) $H_3C-\underset{\underset{OH}{|}}{CH}-CH_3$

(B) $H_3C-CH=CH_2$

(C) $H_3C-\underset{\underset{Cl}{|}}{CH}-CH_3$

(D) All of these

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Paragraph for Question Nos. 15 to 16

Read the paragraph carefully and answer the following questions:

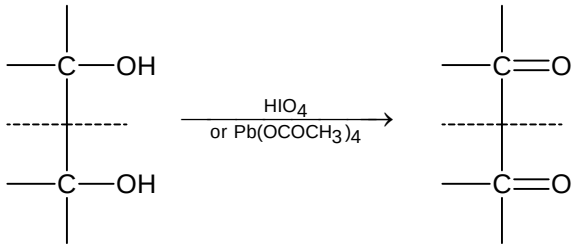
In a polyhydric alcohol, the total numbers of hydroxyl groups are estimated by the following methods:

(i) A known mass of the acetyl derivative of polyhydric alcohol is taken. It is mixed with an excess volume of a standard alkali solution. Both are refluxed. Alkali is used in the hydrolysis of acetyl derivative. The unused alkali is measured by carrying out titration with a standard acid solution and total number of hydroxyl groups (n) can thus be calculated stoichiometrically.

$$A(OH)_n \xrightarrow{CH_3COCl} A(OCOCH_3)_n \xrightarrow{nKOH} A(OH)_n + nCH_3COOK$$

Polyhydric alcohol — Acetyl derivative

(ii) Total number of –OH groups in vicinal position can be calculated by oxidation with periodic acid or lead tetra acetate.



15. 0.436 g of acetyl derivative of a polyhydric alcohol (molecular mass = 92) require 0.336 g KOH for hydrolysis. The total numbers of hydroxyl group(s) in the alcohol are
(A) 1 (B) 2
(C) 3 (D) 4

16. The product formed by the reaction of ethylene glycol and periodic acid (HIO_4) is
(A) acetic acid (B) formic acid
(C) formaldehyde (D) glyoxal

Space for rough work

(Matching list Type)

This section contains **4 multiple choice questions**. Each question has matching lists. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

1. Match the following:

Column – I (Substrate + reagent)		**Column - II (Major product)**	
(P)	$CH_3–CH_2–CH_2–Cl \xrightarrow[C_2H_5OH]{C_2H_5ONa}$	(1)	$CH_3–CH_2–CH=CH_2$
(Q)	$CH_3–\underset{\displaystyle Cl}{\overset{\displaystyle CH_3}{\overset{\displaystyle |}{\underset{\displaystyle |}{C}}}}–CH_3 \xrightarrow[C_2H_5OH]{C_2H_5ONa}$	(2)	$CH_3–CH=CH–CH_3$
(R)	$CH_3-CH_2-\underset{\displaystyle Cl}{\overset{\displaystyle |}{CH}}-CH_3 \xrightarrow[BuOH]{t-BuO^-K^+}$	(3)	$CH_3–CH=CH_2$
(S)	$CH_3-CH_2-\underset{\displaystyle Cl}{\overset{\displaystyle |}{CH}}-CH_3 \xrightarrow[C_2H_5OH]{C_2H_5ONa}$	(4)	$CH_3–\underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}}=CH_2$

Codes:

	P	**Q**	**R**	**S**
(A)	3	4	1	2
(B)	1	4	3	2
(C)	3	2	1	4
(D)	4	3	2	1

Space for rough work

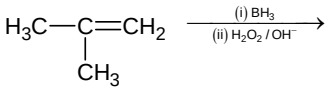
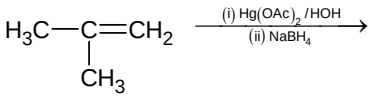
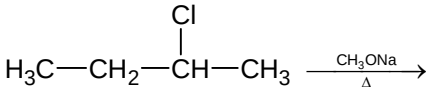
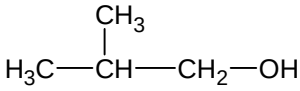
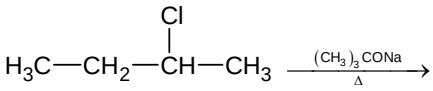
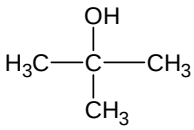
2. Match the following:

Column I		Column II	
(P)	Hell–Volhard-Zelinsky reaction	(1)	Preparation of amines
(Q)	Reformatsky reaction	(2)	Preparation of β–hydroxy ester
(R)	Claisen condensation reaction	(3)	α–halogenation of carboxylic acid
(S)	Schmidt reaction	(4)	Preparation of β–keto ester

Codes:

	P	Q	R	S
(A)	4	1	2	3
(B)	1	4	3	2
(C)	3	2	1	4
(D)	3	2	4	1

3. Match the Column – I with Column – II:

Column – I		Column – II	
(P)		(1)	$CH_3 - CH_2 - CH = CH_2$
(Q)		(2)	$CH_3 - CH = CH - CH_3$
(R)		(3)	
(S)		(4)	

Codes:

	P	Q	R	S
(A)	4	1	2	3
(B)	3	4	2	1
(C)	3	2	1	4
(D)	3	2	4	1

Space for rough work

4. Match the following:

Column – I		Column – II	
(P)	$RCONH_2 \xrightarrow{Br_2/KOH} RNH_2$	(1)	Curtius rearrangement
(Q)	$RCOOH \xrightarrow[H_2SO_4]{N_3H} R-NH_2$	(2)	Lossen rearrangement
(R)	$RCOCl \xrightarrow[\Delta]{NaN_3} R-NH_2$	(3)	Schmidt rearrangement
(S)	$R-\overset{O}{\overset{\|}{C}}-Cl \xrightarrow[OH^-/\Delta]{(i)\ NH_2OH}$	(4)	Hoffmann rearrangement

Codes:

	P	Q	R	S
(A)	4	1	2	3
(B)	1	4	3	2
(C)	3	2	1	4
(D)	4	3	1	2

Space for rough work

Mathematics

PART – III

SECTION - A

Straight Objective Type

This section contains **10 multiple** choice questions numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. If $X = \{8^n - 7n - 1 : n \in N\}$ and $Y = \{49(n-1) : n \in N\}$, then
   (A) $X \subseteq Y$ (B) $Y \subseteq X$
   (C) $X = Y$ (D) none of these.

2. $\lim\limits_{x\to 1}\left(\sec\frac{\pi}{2^x}\right)(\ln x)$ is equal to
   (A) $-\frac{\pi}{2\ln 2}$ (B) $\frac{\pi}{2}$
   (C) $\frac{\pi}{2\ln 2}$ (D) $\frac{2}{\pi\ln 2}$.

3. If $f\left(x+\frac{y}{8}, x-\frac{y}{8}\right) = xy$ then $f(m,n) + f(n,m) = 0$
   (A) only when m = n (B) only when $m \neq n$
   (C) only when m = – n (D) for all m & n.

4. If $\int \frac{2\cos x - \sin x + \lambda}{\cos x + \sin x - 2}dx = A\ln|\cos x + \sin x - 2| + Bx + C$ then the ordered pair triplet A, B, C is
   (A) $\left(\frac{1}{2}, \frac{3}{2}, -1\right)$ (B) $\left(\frac{3}{2}, \frac{1}{2}, -1\right)$
   (C) $\left(\frac{1}{2}, -1, \frac{-3}{2}\right)$ (D) $\left(\frac{3}{2}, -1, \frac{1}{2}\right)$.

Space for rough work

5. $I_1 = \int_0^1 e^{-x}\cos^2 x\,dx \quad I_2 = \int_0^1 e^{-x^2}\cos^2 x\,dx \quad I_3 = \int_0^1 e^{-\frac{x^2}{2}}\cos^2 x\,dx \quad I_4 = \int_0^1 e^{-\frac{x^2}{2}}dx$ then

(A) $I_2 > I_4 > I_1 > I_3$ (B) $I_2 < I_4 < I_1 < I_3$

(C) $I_1 < I_2 < I_3 < I_4$ (D) $I_1 > I_2 > I_3 > I_4$.

6. $f(x)$ is a differentiable function satisfying the relation $f(x) = x^2 + \int_0^x e^{-t} f(x-t)\,dt$, then $\sum_{k=1}^{9} f(k)$ equals

(A) 1060 (B) 1260

(C) 960 (D) 1224

7. If $I = \int \frac{dx}{\sin\left(x-\frac{\pi}{3}\right)\cos x}$, then I equals

(A) $2\log\left|\sin x + \sin\left(x-\frac{\pi}{3}\right)\right| + C$ (B) $2\log\left|\sin\left(x-\frac{\pi}{3}\right)\sec x\right| + C$

(C) $2\log\left|\sin x - \sin\left(x-\frac{\pi}{3}\right)\right| + C$ (D) None of these

8. Consider a function f defined on the set of all non-negative integers such that f(0)=1,f(1)=0and f(n)+f(n-1)=n f(n-1)+(n-1)f(n-2)for $n \geq 2$. Then f(5) =

(A) 40 (B) 44

(C) 45 (D) 60

9. Let $f: R \to R$ be a function such that $f\left(\frac{x+y}{3}\right) = \frac{f(x)+f(y)}{3}, f(0) = 0$ and $f'(0) = 3$, then

(A) $\frac{f(x)}{x^2}$ is differentiable in R

(B) $f(x)$ is continuous but not differentiable in R

(C) f (x) is continuous in R

(D) f(x) is bounded in R

10. Let $f(x) = \frac{\tan x}{x}$, then $\log_e\left(\lim_{x\to 0}\left([f(x)] + x^2\right)^{\frac{1}{\{f(x)\}}}\right)$ is equal, (where [·] denotes greatest integer function and {·} fractional part)

(A) 1 (B) 2

(C) 3 (D) 4

Space for rough work

Comprehension Type

This section contains **3 groups** of questions. Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Question Nos. 11 to 12

Read the following write up carefully and answer the following questions:

Definite integral of any discontinuous non differentiable function is normally solved by property $\int_a^b f(x)\,dx = \int_a^c f(x)dx + \int_c^b f(x)\,dx$; where c $\in$ (a, b) is a point of discontinuity or non differentiability

11. The value of $A = \int_1^{\infty} \left[\operatorname{cosec}^{-1} x\right] dx$ {where [.] denotes greatest integer function} is equal to
    (A) cosec 1 – 1 (B) 1
    (C) 1 – sin 1 (D) none of these.

12. The value of $B = \int_1^{100} \left[\sec^{-1} x\right] dx$ {where [.] denotes greatest integer function} is equal to
    (A) sec 1 (B) 100 – sec 1
    (C) 99 – sec 1 (D) none of these.

Paragraph for Question Nos. 13 to 14

Read the following write up carefully and answer the following questions:

If f : $(0, \infty) \to (0, \infty)$ satisfy $f(x(f(y))) = x^2 y^a$ ($a \in R$), then

13. Value of a is
    (A) 4 (B) 2
    (C) $\sqrt{2}$ (D) 1.

14. $\sum_{r=1}^{n} f(r) \cdot {}^nC_r$ is
    (A) $n2^{n-1}$ (B) $n(n-1)2^{n-2}$
    (C) $n2^{n-1} + n(n-1)2^{n-2}$ (D) 0.

Space for rough work

Paragraph for Question Nos. 15 to 16

Read the following write up carefully and answer the following questions:
Left hand derivative and right hand derivative of a function f(x) at a point x = a are defined as

$$f'(a^-) = \lim_{h\to 0^-} \frac{f(a)-f(a-h)}{h} = \lim_{h\to 0^-} \frac{f(a+h)-f(a)}{h} \text{ and}$$

$$f'(a^+) = \lim_{h\to 0^-} \frac{f(a+h)-f(a)}{h} = \lim_{h\to 0^-} \frac{f(a)-f(a-h)}{h} = \lim_{x\to a^+} \frac{f(a)-f(x)}{a-x} \text{ respectively.}$$

15. If f is odd, which of the following is left hand derivative of at x = – a

(A) $\lim_{h\to 0^-} \frac{f(a-h)-f(a)}{-h}$ (B) $\lim_{h\to 0^-} \frac{f(h-a)-f(a)}{h}$

(C) $\lim_{h\to 0^+} \frac{f(a)+f(a-h)}{-h}$ (D) $\lim_{h\to 0^-} \frac{f(-a)-f(-a-h)}{-h}$

16. If f is even which of following is right hand derivative of f′ at x = a

(A) $\lim_{h\to 0} \frac{f'(a)+f'(-a+h)}{h}$ (B) $\lim_{h\to 0^+} \frac{f'(a)+f'(-a-h)}{h}$

(C) $\lim_{h\to 0^-} \frac{-f'(-a)+f'(-a-h)}{h}$ (D) $\lim_{h\to 0} \frac{f'(a)+f'(-a+h)}{-h}$

(Matching list Type)

This section contains **4 multiple choice questions**. Each question has matching lists. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

17. Now match the following List–I with List–II.

	List – I		List – II
(P)	$f'(x)\sqrt{3x^2+6}$ and y = f(x^3) then at x = 1, $\frac{dy}{dx}$ =	1.	– 2
(Q)	If f be a differentiable function such that f(xy) = f(x) + f(y) $\forall$ x $\in$ R, y $\in$ R then f(e) + f(1/e) =	2.	0
(R)	If f be a twice differentiable function such that f″(x) = – f(x) and f′(x) = g(x) if h(x) = $(f(x))^2$ + $(g(x))^2$and h(5) = 9 then h(10) = A find value of 3A	3.	9
(S)	y = $\tan^{-1}(\cot x) + \cot^{-1}(\tan x)$ $\frac{\pi}{2} < x < \pi$ then $\frac{dy}{dx}$	4.	27

Codes:

	P	Q	R	S
(A)	4	3	2	1
(B)	1	2	3	4
(C)	3	2	4	1
(D)	4	1	3	2

Space for rough work

18. Now match the following List–I with List–II.

	List – I		List – II
(P)	Number of points of discontinuity of $f(x) = \tan^2 x - \sec^2 x$ in $(0, 2\pi)$ is	1.	3
(Q)	Number of points at which $f(x) = \sin^{-1}x + \tan^{-1}x + \cot^{-1}x$ is non differentiable	2.	0
(R)	Number of points of discontinuity of $y = [\sin x]$, $x \in [0, 2\pi)$ is N then value of |2N| where [.] represents greatest integer function	3.	2
(S)	Number of points where $y = \|(x-1)^3\| + \|(x-2)^5\| + \|x-3\|$ is non differentiable	4.	4

Codes:

	P	Q	R	S
(A)	4	3	2	1
(B)	1	2	3	4
(C)	4	1	3	2
(D)	3	2	4	1

19. Now match the following List–I with List–II.

	List – I		List – II
(P)	Let $f : R \to R$ be a differentiable function and $f(1) = 1$, $f'(1) = 3$; then the value of $\lim\limits_{x\to 1} \int_1^{x^2} \frac{f(t)-t}{(x-1)^2} dt$	1.	0
(Q)	$\lim\limits_{n\to\infty} \left(\frac{1+\sqrt[n]{4}}{2}\right)^n$ is equal to	2.	– 1
(R)	If $\lim\limits_{n\to\infty} \frac{2x}{\pi}\tan^{-1}(nx)$; $x > 0$ then $\lim\limits_{x\to 0^+}\left[f(x)-1\right]$ is {where [.] represents greatest integer function.	3.	2
(S)	$\lim\limits_{n\to\infty}\left[\sum\limits_{r=1}^{n}\frac{1}{2^r}\right]$ where [.] denotes greatest integer function	4.	4

Codes:

	P	Q	R	S
(A)	4	3	2	1
(B)	1	2	3	4
(C)	2	3	4	1
(D)	4	1	2	3

Space for rough work

20. Now match the following List–I with List–II.

	List – I		List – II
(P)	$\int_4^{10} \frac{\left[x^2\right]dx}{\left[x^2-28x+196\right]+\left[x^2\right]} =$	1.	$\frac{1}{100}$
(Q)	$\int_{-1}^{2} \frac{\lvert x\rvert}{x} dx =$	2.	3
(R)	$\lim_{n\to\infty} \frac{1^{99}+2^{99}+3^{99}....n^{99}}{n^{100}} = 2A$ then value of A	3.	1
(S)	$5050\int_{-1}^{1} \sqrt{x^{200}}dx = \frac{1}{\alpha}$ then $\alpha =$	4.	$\frac{1}{200}$

Codes:

	P	Q	R	S
(A)	4	3	2	1
(B)	1	2	3	4
(C)	2	3	4	1
(D)	4	1	2	3

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